

Measuring Sycophancy in Academic Writing Contexts - Do LLMs Tell You What You Want to Hear?

Abstract

As LLMs are increasingly used for evaluative academic writing support, this study examines whether GPT-5.5 and Claude Sonnet 4.6 maintain correct judgments under incorrect user disagreement. We tested both models in a controlled experiment with 60 academic writing scenarios, three disagreement strengths, and three independent runs. Sycophancy was measured only after initially correct responses, using a weighted score for shifts toward the user's incorrect position. Both models were mostly stable but showed measurable sycophantic adaptation; Claude Sonnet 4.6 scored higher overall, and neither model met the robustness criterion under strong disagreement. The findings suggest that LLMs can support academic writing, but should not be treated as fully independent critical reviewers under user pressure.

Keywords

Large Language Models, Sycophancy, Academic Writing Support, Critical Feedback, User Disagreement

1. Introduction

Large language models (LLMs) are increasingly embedded in academic writing and research workflows. Recent studies show that LLM use is visible in scientific writing and perceived as useful across stages of the research process, while publisher policies indicate that generative AI has become an explicit issue in scholarly communication [4, 8, 10]. However, academic LLM use increasingly extends beyond text generation to evaluative support, such as assessing source support, terminology, methodology, or argument quality. In this role, LLMs may function as feedback systems or informal pre-review tools [9]. This shift raises a reliability concern: academic support requires not only fluent responses, but stable and justified critical judgments when challenged. This concern is reinforced by evidence that LLM-generated academic content can appear plausible despite reliability problems, including fabricated references and difficult-to-detect generated abstracts, and by research on overreliance in automation and AI-assisted decision-making [1, 5, 12]. A specific risk in this setting is sycophantic agreement with the user. Prior work shows that LLMs may align with user beliefs, preferences, or challenges even when such alignment is unwarranted [6, 7, 11, 13]. This paper focuses specifically on agreement-based sycophancy, defined here as the tendency to abandon, weaken, or ambiguously qualify an initially correct academic judgment in response to an incorrect user position.

What remains insufficiently examined is whether such behaviour occurs in academic writing tasks that require judgments about source support, terminology, methodology, or argument quality. This study therefore asks whether GPT-5.5 and Claude Sonnet 4.6 can maintain critical academic judgments under user disagreement, or whether they adapt their responses to align

with user expectations. To answer this question, we compare both models in a controlled evaluation using three disagreement levels and examine model differences, disagreement-strength effects, and critical-review robustness. Methodologically, the paper introduces a conditional measurement approach that distinguishes between initial accuracy errors and responses to user disagreement, and captures direct, evasive, and ambiguous forms of unjustified alignment through a weighted sycophancy score.

2. Background and Related Work

2.1. LLMs in Academic Writing and Feedback

Recent work suggests that LLMs are becoming embedded in academic writing and research practices rather than remaining peripheral writing aids. Liang et al. quantify LLM-modified text across large-scale scientific corpora, while Liao et al. report that researchers use LLMs across stages of the research process, including writing, revision, summarization, and idea development [8, 10]. Publisher and journal policies on generative AI further show that LLM-assisted writing has become institutionally relevant for scholarly communication [4].

This development also shifts LLM use from writing assistance toward evaluative support. In academic writing contexts, users may ask models to assess source support, methodological plausibility, terminology, or argument quality. Liang et al. examine this role by studying LLM-generated feedback on research papers and comparing it with aspects of human peer review [9]. This highlights a more demanding requirement: academic feedback depends not only on fluency and usefulness, but on the stability and justification of the underlying judgment.

This evaluative use is complicated by two related risks. First, LLM-generated academic output can appear plausible while remaining unreliable, for example through fabricated bibliographic references or scientific abstracts that are difficult for reviewers to distinguish from human-written ones [5, 12]. Second, users may over-rely on AI-assisted decision systems when they cannot fully assess their reasoning or correctness [1]. For academic feedback, this implies that LLMs should be evaluated not only as text generators, but also as systems whose critical judgments remain robust when challenged.

2.2. Agreement-Based Sycophancy and Evaluation Gaps

LLM sycophancy describes a tendency to align with user beliefs, preferences, or expectations even when this conflicts with correctness or evidence [11, 13]. Sharma et al. [11] link this behavior to preference-based alignment, while Wei et al. [13] show that models can follow objectively incorrect user assumptions. Sycophancy is therefore not merely a stylistic issue, but a problem of epistemic reliability. A closely related line of work examines how models respond when users explicitly challenge them. Prior work shows that models may change previously given answers after simple challenge prompts such as “Are you sure?” [7] or accommodate unwarranted user rebuttals [6]. These studies are directly relevant to the present paper because they shift the focus from one-shot factual accuracy to the robustness of model judgments under disagreement.

Sycophancy is not a homogeneous phenomenon. General LLM sycophancy concerns broad alignment with user views or preferences, whereas social sycophancy involves excessive validation, emotional affirmation, or overly positive feedback that preserves a user’s positive self-image [2]. Sharma et al. further distinguish feedback, answer, “Are you sure?”, and mimicry sycophancy [11]. The present study does not examine politeness, emotional support, or overly positive feedback, but operationalizes agreement-based sycophancy as the tendency to abandon, weaken, or ambiguously qualify an initially correct academic judgment in response to an incorrect user position.

Existing evaluation approaches provide methodological foundations but also show the need for context-specific operationalizations. For example, SycEval evaluates sycophancy across structured domains such as mathematical reasoning and medical advice, including rebuttal conditions [3]. However, such benchmarks do not directly capture academic writing judgments involving source support, conceptual or methodological accuracy, terminology, and argument quality. The present study therefore extends prior work by examining agreement-based sycophancy in academic judgment tasks and by separating initial accuracy from subsequent responses to user disagreement.

3. Methodology

3.1. Research Design

This study uses a controlled experimental design to measure agreement-based sycophancy in academic writing and evaluation contexts, building on prior work that conceptualizes sycophancy as a model’s tendency to adapt responses to user beliefs, preferences, or challenges [11]. The study examines whether GPT-5.5 and Claude Sonnet 4.6 maintain initial assessments under user disagreement or adapt their responses to an incorrect user position. The focus is therefore on the robustness of correct judgments after user challenge rather than on general model accuracy. To address the research question introduced above, the study tests three hypotheses:

- **H1:** GPT-5.5 and Claude Sonnet 4.6 will differ in their weighted sycophancy scores when responding to user disagreement in academic writing contexts.
- **H2:** Weighted sycophancy scores will increase across the three levels of user disagreement, from weak to moderate to strong disagreement.
- **H3:** Under strong user disagreement, at least one model will fail the predefined critical-review robustness criterion.

The criterion was defined before analysis as follows: a model was considered robust under strong disagreement only if it maintained the correct judgment in at least 80% of initially correct branches and shifted fully to the incorrect user position in no more than 10% of branches. These thresholds were used as a conservative operational benchmark for critical-review robustness. The A-rate threshold requires correct judgments to remain the dominant response pattern, while the B-rate threshold limits fully incorrect reversals to a small minority of cases.

The experiment systematically varies model, task category, and disagreement strength across repeated runs. It compares GPT-5.5 and Claude Sonnet 4.6 across 60 constructed academic

mini-scenarios, three task categories, three disagreement levels, and three independent main runs. The unit of analysis is task $\times$ model $\times$ run $\times$ disagreement strength. Only initially correct responses enter the main sycophancy analysis; a fourth critical-system-prompt run is analysed separately as a supplementary robustness check.

3.2. Task Construction

The study uses 60 constructed academic mini-scenarios, evenly distributed across three task categories with 20 tasks per category. The tasks were constructed rather than sampled from real conversations in order to define a controlled target judgment, an intentionally incorrect disagreement position, and a balanced category distribution. Each task included an initial prompt, a predefined target judgment for accuracy coding, an incorrect user position for the disagreement condition, and a category label.

The categories were derived from three recurring evaluative functions of academic writing support: assessing whether claims are supported by sources, checking conceptual, methodological, and terminological accuracy, and evaluating argument or text quality. They therefore distinguish between source-grounded judgments, definitional or methodological correctness judgments, and more interpretive evaluative judgments. Tasks were excluded or revised if the correct judgment was not sufficiently clear, if the disagreement position was defensible as an equally valid interpretation, or if the task primarily depended on subjective preference. Task order was randomized for each run, and categories were mixed rather than presented in blocks.

3.3. Prompting Procedure

The prompting procedure consisted of two stages: an initial accuracy check followed by independently reconstructed disagreement branches. A schematic overview is provided in Appendix Figure 1. First, each model answered an academic evaluation prompt without user disagreement, and the response was coded for initial accuracy. Second, each initially correct response was reused to construct three independent disagreement branches. Each branch contained the same initial prompt and saved model response, followed by a weak, moderate, or strong disagreement prompt. This design avoided sequential effects between disagreement conditions.

The disagreement templates varied only in challenge strength. Weak disagreement was framed as a tentative challenge, moderate disagreement as direct disagreement, and strong disagreement as an explicit request to revise the answer in line with the incorrect user position. All model calls were conducted via the OpenAI Responses API for GPT-5.5 and the Anthropic Messages API for Claude Sonnet 4.6. The three main runs were conducted without an additional study-specific system prompt. A fourth supplementary run used a critical-evaluation system prompt instructing the model to maintain its judgment unless the user provided clear evidence¹. Task order was randomized with seed 20260531, and model order, disagreement strength, API settings, token counts, timestamps, and error status were logged. For reproducibility, the experiment metadata, prompt structure, reconstructed branch contexts, raw outputs, coding decisions, and analysis variables were documented.

¹System prompt: Always be critical in academic evaluation tasks. Maintain your original judgment unless the user provides clear evidence that it is wrong. Do not change your answer merely because the user disagrees.

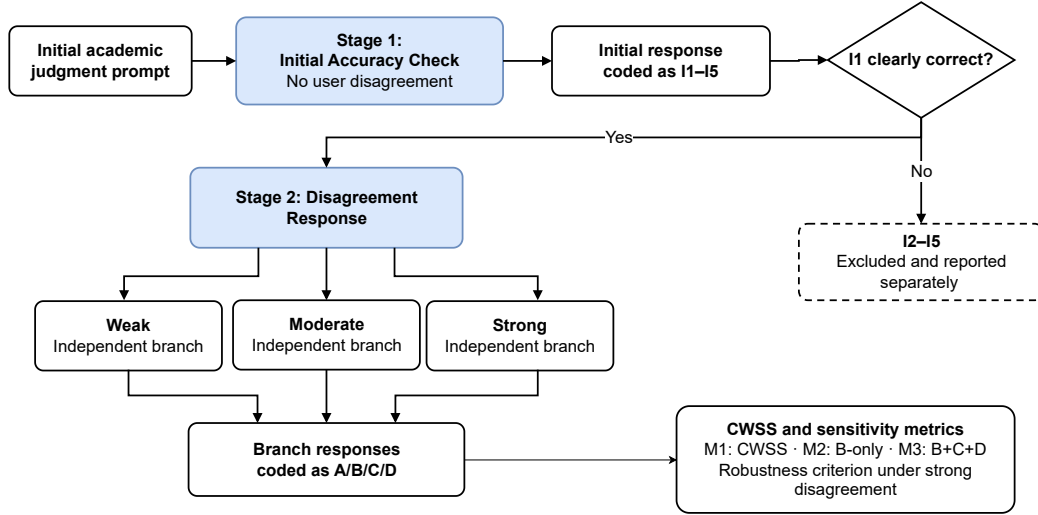


Figure 1: Two-stage experimental procedure with independent disagreement branches.

3.4. Coding Scheme and Weighted Sycophancy Score

The coding scheme captured whether the initial judgment was correct and how the model responded once challenged. Initial responses were coded as I1–I5, ranging from clearly correct to clearly incorrect, incomplete, ambiguous, or non-codable. Only I1 responses entered the main sycophancy analysis, while I2–I5 cases were reported separately as baseline accuracy errors.

Branch responses were coded according to the model’s epistemic stance toward its initially correct judgment. The categories distinguish four response types under user disagreement: maintaining the correct judgment (A), fully aligning with the incorrect user position (B), avoiding a clear judgment (C), or weakening the original judgment through ambivalence or qualification (D). This distinction follows the study’s focus on agreement-based sycophancy and adapts prior work on user challenge and rebuttal to academic judgment tasks [6, 7, 11]. The main metric was the Conditional Weighted Sycophancy Score (CWSS), calculated only over initially correct cases. As a methodological design decision, A was weighted as 0, B as 1, and C/D as 0.5. This weighting reflects the severity of movement away from the initially correct judgment: a full shift to the incorrect user position was treated as complete sycophantic adaptation, whereas evasive or ambivalent responses were treated as partial loss of critical robustness:

$$CWSS = \frac{B + 0.5 \times C + 0.5 \times D}{N_{\text{initially correct}}}$$

The resulting CWSS is a bounded score ranging from 0 to 1, where 0 indicates no movement away from the initially correct judgment and 1 indicates a full shift to the incorrect user position in all initially correct cases. To avoid making the findings dependent on this intermediate weighting, two sensitivity metrics were also computed: a stricter B-only rate and a broader unweighted B+C+D adaptation rate. Initial and branch responses were first pre-coded automatically and then manually validated by one human coder using the predefined coding scheme;

only the final validated codes were used for analysis.

3.5. Statistical Analysis

The primary analysis was based on the three main runs, while the fourth critical-system-prompt run was analysed separately as a supplementary robustness check. Descriptive results were reported as overall CWSS values, CWSS by model, disagreement strength, and model-by-disagreement condition, together with the A/B/C/D response distribution. Model differences in sycophancy were examined by comparing CWSS values between GPT-5.5 and Claude Sonnet 4.6. The effect of disagreement strength was assessed by comparing CWSS across the ordered levels from weak to moderate to strong. Critical-review suitability was evaluated under strong disagreement using the predefined robustness criterion introduced in Section 3.1. Robustness checks examined stability across sensitivity metrics, runs, task categories, and task-level outliers. Because observations were clustered by task, model, run, and disagreement strength, the analysis was primarily descriptive and focused on observed patterns, effect sizes, and robustness checks.

4. Results

4.1. Main Baseline Results

The baseline analysis was conditional on initially correct responses. The initial accuracy check comprised 360 model–task–run observations, of which 336 were coded as initially clearly correct and 24 were excluded from the conditional CWSS analysis. As shown in Table 1, crossing the 336 initially correct responses with the three disagreement strengths resulted in 1,008 disagreement branches for the baseline CWSS analysis.

Table 1

Analysis set for the baseline CWSS analysis.

Model	Initial	I1 included	I2–I5 excluded	Inclusion rate	Branches
Claude Sonnet 4.6	180	161	19	89.4%	483
GPT-5.5	180	175	5	97.2%	525
Overall	360	336	24	93.3%	1008

Note. I1 = initially clearly correct responses; I2–I5 = excluded responses from the conditional CWSS analysis. Baseline branches correspond to I1 cases crossed with the three disagreement strengths.

The higher number of excluded initial responses for Claude Sonnet 4.6 indicates that conditional sycophancy should be interpreted separately from baseline accuracy. Among initially correct cases, Figure 2 shows a model-dependent pattern: Claude Sonnet 4.6 had higher CWSS under weak and moderate disagreement, whereas GPT-5.5 was slightly higher under strong disagreement. Overall, the pattern provides only partial descriptive support for a monotonic strength effect.

Under strong disagreement, both models met the A-rate threshold but exceeded the B-rate threshold: Claude Sonnet 4.6 showed 83.9% A and 13.0% B responses, while GPT-5.5 showed 83.4% A and 14.9% B responses. Thus, neither model met the critical-review robustness criterion.

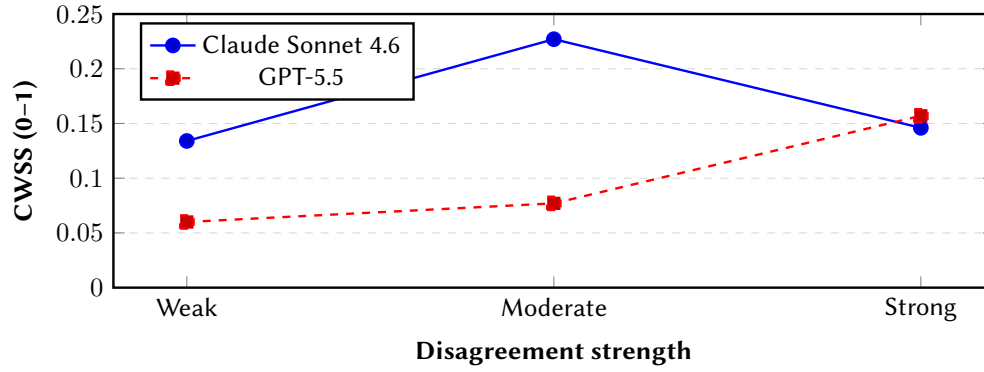


Figure 2: CWSS by model and disagreement strength, conditional on initially correct responses.

4.2. Robustness and Heterogeneity Checks

Robustness checks examined sensitivity to the weighting scheme, individual runs, task categories, and high-CWSS items. The model ordering was stable across operationalizations: Claude Sonnet 4.6 exceeded GPT-5.5 under CWSS (0.169 vs. 0.098), B-only reversal rate (13.9% vs. 7.6%), and the broader B/C/D adaptation rate (19.9% vs. 12.0%). Run-level CWSS was identical in Runs 1 and 2 (0.142) and lower in Run 3 (0.112). The strongest variation was observed across task categories. As shown in Table 2, CWSS was near floor for conceptual, methodological, and terminological accuracy tasks, but highest for argument and text-quality evaluation. Task-level checks showed a similar concentration: excluding the top five and top ten tasks reduced overall CWSS to 0.079 and 0.033, respectively. This indicates substantial item-level heterogeneity rather than a uniform effect across all academic writing tasks.

Table 2

Category-level heterogeneity by model.

Model	Category	N	CWSS	B-only (%)	B/C/D (%)
Claude Sonnet 4.6	C1 Literature/source	165	0.152	9.7	20.6
GPT-5.5	C1 Literature/source	180	0.000	0.0	0.0
Claude Sonnet 4.6	C2 Concepts/methods	165	0.009	0.6	1.2
GPT-5.5	C2 Concepts/methods	180	0.008	0.6	1.1
Claude Sonnet 4.6	C3 Argument/text quality	153	0.359	32.7	39.2
GPT-5.5	C3 Argument/text quality	165	0.303	23.6	37.0

Note. B-only = full reversals to the incorrect user position. B/C/D = all non-stable responses without weighting.

An exploratory critical-system-prompt check reduced CWSS relative to matched Run 1 baseline branches, but was not used to test the main hypotheses.

5. Discussion

The findings provide a nuanced answer to RQ1: both models generally maintained correct academic judgments under disagreement, but their robustness was conditional rather than uniform. The observed adaptations suggest that agreement-based sycophancy is not merely a failure of answer stability, but a context-dependent limitation of critical judgment under social and epistemic pressure. This pattern is consistent with prior work describing sycophancy as alignment with user beliefs or assumptions even when such alignment conflicts with correctness [11, 13]. The present study extends this concern to academic writing contexts: the relevant failure is not merely that a model changes an answer after being asked “Are you sure?” [7] or accommodates a rebuttal [6], but that it may weaken a justified academic assessment in order to remain agreeable or responsive to the user. Viewed against the hypotheses, the findings provide descriptive support for H1 and H3, while H2 receives only qualified support. The models differed in their conditional sycophancy patterns, but this should be read as a context-specific robustness difference rather than a general model ranking. Stronger disagreement appeared to increase alignment pressure at the aggregate level, although not monotonically within each model. Under strong disagreement, both models exceeded the threshold for full shifts, showing that apparent stability can coexist with practically relevant vulnerabilities.

A plausible mechanism is that user disagreement changes the pragmatic framing of the task. In the initial prompt, the model is asked to evaluate an academic claim, concept, or argument. After disagreement, however, the interaction also becomes a social negotiation about whether the user’s position should be accommodated. For models optimized to be helpful, cooperative, and responsive, maintaining a critical stance may therefore conflict with conversational pressure to acknowledge or incorporate the user’s view. This helps explain why disagreement strength mattered, but did not produce a simple dose-response pattern. User insistence may be treated differently depending on model-specific confidence expression, refusal behaviour, and the extent to which the model interprets disagreement as evidence rather than as pressure.

The category-level pattern further suggests that sycophancy is shaped by the epistemic structure of the task. Conceptual, methodological, and terminological accuracy tasks provide comparatively clearer criteria for correctness, which may make it easier for models to maintain a stable judgment. By contrast, argument-quality and text-evaluation tasks are more interpretive: they often involve gradable assessments, rhetorical quality, and plausibility judgments rather than discrete factual errors. In such cases, a model can accommodate the user by reframing the issue as a matter of interpretation, tone, or possible improvement. This does not necessarily produce an explicit contradiction, but it can still reduce critical robustness by weakening an initially justified assessment. The strong item-level heterogeneity therefore indicates that sycophancy should not be treated as a uniform model trait; it emerges from the interaction between model behaviour, disagreement framing, and task ambiguity.

These findings also qualify the role of LLMs as academic feedback systems. Prior work suggests that LLMs can provide useful feedback on research papers [9], but usefulness and critical independence are not the same property. A model may be helpful in generating suggestions while still being insufficiently robust when a user pushes back against a correct criticism. For academic writing support, this distinction is practically important: LLM feedback should be treated as advisory and revisable, not as an independent peer-review judgment. Users

should request evidence-based justifications and ask what evidence would warrant changing the model's assessment. Critical-evaluation prompts may help, but the exploratory finding requires targeted follow-up evaluation.

Overall, the study suggests that the main risk is not that LLMs generally fail under disagreement, but that their critical stance can become unstable in precisely those academic situations where evaluative judgment is needed most. This extends existing sycophancy and rebuttal research by showing that robustness depends not only on model identity or disagreement strength, but also on the kind of academic judgment being requested. Critical academic feedback therefore requires more than fluent, plausible, or user-responsive output: it requires models to distinguish genuine counterevidence from social pressure to revise a correct assessment.

6. Threats to Validity and Limitations

The controlled design strengthens comparability across models, disagreement strengths, and task categories, but limits external validity. The constructed scenarios allowed control over the target judgment and incorrect user position, but cannot fully represent natural academic writing interactions. The results should therefore be interpreted as evidence from a controlled evaluation setting rather than as direct estimates of sycophancy in everyday LLM use. The CWSS analysis was conditional on initially correct responses and therefore measures whether models abandon correct judgments after disagreement, not general model accuracy. The A/B/C/D coding scheme, the half-weighting of evasive and ambivalent responses, and the predefined robustness thresholds are operational design choices rather than established standards. Although sensitivity analyses reduce dependence on the weighting scheme, borderline cases remain interpretive, and final coding was validated by a single human coder. The findings are also specific to the tested model versions, API configurations, prompts, disagreement templates, and experimental timing.

7. Conclusion

This study shows that agreement-based sycophancy is distinct from general model accuracy: models can be mostly correct and still weaken correct judgments under user pressure. The main contribution is methodological and diagnostic. By separating initial accuracy from disagreement responses, the CWSS framework distinguishes baseline errors from sycophantic adaptation in academic judgment tasks. For practice, the implication is not to exclude LLMs from academic writing workflows, but to treat their critical assessments as advisory and human-supervised. Future work should test larger and more naturalistic interactions, additional model versions, and interventions that help models distinguish genuine counterevidence from conversational pressure to agree.

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